Venkata Lakshmi Parimala Pasupuleti

Forward Deployed AI Engineer · Python · AWS · Docker · AI Model Integration · Customer-Facing Deployment · Open Source Production AI

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PROFESSIONAL SUMMARY

Forward deployed engineer with 9+ years of production Python and cloud infrastructure experience, specializing in deploying AI systems into complex customer environments and owning the full technical lifecycle from discovery to production. AgentProbe is the core proof point: an open-source production AI platform I built end-to-end that routes inference across multiple large AI model providers (Anthropic, OpenAI, Ollama) via clean API integrations, runs a 5-metric quality engine, and gates regressions in CI before anything ships - exactly the kind of infrastructure that sits between AI models and the customers who rely on them. At Magizhs, I took an agentic AI system from ambiguous customer requirements to production deployment in weeks, owning architecture, integration, and handoff solo. At Entrata, I built and maintained enterprise-grade production systems (AWS, Docker, RabbitMQ, FastAPI) for 9+ years - 99.5% uptime, full PHI compliance, zero security incidents - at the reliability standard enterprise customers expect. I communicate fluently across engineering, product, and scientific stakeholders, and I learn new domains fast. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Deployment & Cloud Infrastructure: Python (9yr), AWS (S3, EC2, RDS, Glue), Docker (containerization, microservices), FastAPI, REST APIs, PostgreSQL, MySQL, Oracle, Redis, RabbitMQ, CI/CD (GitHub Actions), Cloud-Native Architecture, Linux, SSH, Git

AI Model Integration & Evaluation: LLM API Integration (Anthropic, OpenAI, Ollama), Multi-Provider Model Routing, LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, LLM Evaluation, LLM-as-Judge, Hallucination Detection, RAG Pipelines, ChromaDB, Embeddings, Prompt Engineering, MCP, HITL, Guardrails, OCR, BERT, spaCy, NLP, PyTorch, scikit-learn

Enterprise Reliability & Customer Delivery: Production Observability & Monitoring, CI-Integrated Regression Gates, Testing Strategy (Unit, Integration, End-to-End), Performance Engineering, PHI/HIPAA Compliance, Data Governance, Security, Cross-Functional Stakeholder Delivery, Technical Documentation, Agile/Scrum, TypeScript, React.js

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Customer-Facing Deployment

- Led end-to-end deployment of a production agentic AI system (LangGraph, FastAPI, Python) for an external customer - from technical discovery through production implementation; gathered requirements from scientific and engineering stakeholders, architected the integration, owned testing and observability, and delivered a working system in weeks with full autonomy. Served as the sole technical contact through handoff.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and system design; translated complex research outputs for academic and non-technical audiences.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Enterprise-grade production infrastructure at the reliability standard external customers require** - AWS S3/EC2/RDS/Glue; Docker containerization (50% scalability improvement, 30% deployment time reduction); RabbitMQ async processing with retry logic, idempotency guarantees, and dead-letter handling; Redis caching; 99.5% uptime across multi-tenant SaaS; 100% PHI compliance with Libsodium-encrypted pipelines; zero security incidents across 7yr of production operation.
- **Production AI-adjacent systems and data pipelines** - owned HUD income-limit pipeline for 7 years (federal data → 180,000 domain records powering eligibility decisions for 100K+ housing units); 200+ SQL query optimizations (30% performance gain); 20+ customer-facing production modules shipped end-to-end across leasing, compliance, and document workflows.
- **Cross-functional delivery and stakeholder communication** - 120+ Agile sprints with product, compliance, and operations stakeholders; built internal debugging tools reducing triage time by 40%; 300+ peer code reviews; GitHub Actions CI/CD; mentored 5+ engineers; leveraged Claude Code to accelerate delivery.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: Open-Source Multi-Provider AI Model Platform with CI Regression Gate

AI Model Integration · Open Source · Production · Customer-Ready

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production platform I built end-to-end that integrates with large AI models through clean API abstractions - routes inference across Anthropic, OpenAI, and Ollama via a unified provider interface; 15-endpoint FastAPI service; 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate); CI-integrated regression gate that blocks quality degradation before it ships to customers. Zero regressions across 1,240+ test cases. Open-source and deployable in enterprise environments.

Leasel: Agentic AI Platform - Concept to Production Deployment

Production AI · End-to-End · Customer-Facing

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Three-surface production AI platform built solo from discovery through deployment - document intelligence (OCR, section-aware chunking, 100% citation correctness), 8-agent autonomous investigation engine (tool routing, memory, HITL escalation), and customer self-service with deterministic guardrails. Full testing, observability, and CI integration - 93.6% precision, 87.9% F1 across 1,240 test cases. Enterprise-ready reliability standards throughout.

AI on FHIR: Full-Stack Healthcare AI - HIPAA-Compliant Production

Healthcare · APIs · HIPAA · Full-Stack

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack customer-facing AI system integrating with healthcare FHIR APIs via NLP (BERT embeddings, LLM inference) - deployed with HIPAA-compliant RBAC, authentication, authorization, and full data isolation; reduced retrieval time ~60%. Demonstrates both life sciences data fluency (FHIR standard) and enterprise security and data governance standards.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017